

COMP1017 (Mathematics for Computer Scientists) Test 2

Academic Session 2017-2018, Autumn Semester

There are TWO (2) questions pertaining to sets and functions on ONE (1) page. Please answer ALL the questions. Time allowed 30 minutes.

1. There are 400 students in a college. Each of the students is required to read several different newspapers. Let S denote the set of students in the college, and T the set of newspapers that are provided to all the students, respectively.

(a) Determine $|S|$, $|P(S)|$, and $|S \times T|$. Show your work (you do not have to give the final figures).

[15 marks]

(b) Assume that each newspaper is read by exactly 80 students. Let A and B represent the set of students who have read the newspapers 'a' and 'b', respectively. Determine the range of $|A \cup B|$. Show your work.

[16 marks]

(c) Assume that there are 8 different newspapers in total. Define a function $f: S \rightarrow P(T)$ such that $f(x) = t$, where the cardinality $|t| = 5$. Is f injective? Is f surjective? Explain your answers.

[19 marks]

2. In the questions below, let both f and g be functions.

(a) Define $f, g: \mathbb{R} \rightarrow \mathbb{Z}$, such that $f = \lfloor x^2 \rfloor$, and $g = \left\lceil \frac{x+1}{2} \right\rceil$. Determine $f \circ g(-1.5)$ and $g \circ f(-1.5)$, respectively. Show your work.

[16 marks]

(b) Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d\}$. Define $f: A \rightarrow B$ and $g: B \rightarrow A$ such that $f(1) = a, f(2) = d, f(3) = c, f(4) = b$, and $g(a) = 3, g(b) = 2, g(c) = 4, g(d) = 1$, respectively. Determine $(f \circ g)^{-1}$. Show your work.

[16 marks]

(c) The sets A and B are defined as in (b). Let B_1 and B_2 be two subsets of B respectively. Define $S = \{f \mid f: A \rightarrow B_1\}$ the set of all functions mapping A to B_1 , and $T = \{g \mid g: A \rightarrow B_2\}$ the set of all functions mapping A to B_2 . Show that if $B_1 \subseteq B_2$ then $S \subseteq T$.

[18 marks]